



MATHEMATICAL PHYSICS

SEMESTER 1, JANUARY EXAM
2018–2019

MP363 Quantum Mechanics I

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Time allowed: 2 hours

Answer **ALL** questions

1. Consider a particle of mass m on an infinite one-dimensional line.

- (a) If the position operator $\hat{x}$ is applied to a wavefunction $\phi(x)$, what is the result?

If the momentum operator $\hat{p}$ is applied to $\phi(x)$, what is the result?

[3 marks]

- (b) Find out if the functions e^{ikx} and $\sin(kx)$ are eigenfunctions of the momentum operator. If so, find the corresponding eigenvalues.

[6 marks]

- (c) Consider a normalized wavefunction $\phi_0(x)$. The expectation value of momentum in this state is p_0 . Find the expectation value of momentum in the state

$$\phi_{\text{new}}(x) = e^{iqx/\hbar} \phi_0(x) .$$

Based on your result, explain the physical significance of multiplying a wavefunction by the phase factor $e^{iqx/\hbar}$.

[9 marks]

- (d) Consider the time-independent wavefunction $\psi(x) = Ne^{-|x|/\lambda}$, i.e.,

$$\psi(x) = \begin{cases} Ne^{-x/\lambda} & \text{for } x \geq 0 \\ Ne^{+x/\lambda} & \text{for } x < 0 \end{cases}$$

where λ and N are real positive constants.

Calculate N in terms of λ so that the wavefunction is normalized.

If the state of the particle is $\psi(x)$, calculate the probability of finding the particle in the region $x \in [-\lambda, \lambda]$.

[14 marks]

2. Consider a spin-1/2 system. The components of the spin are described by the operators

$$\hat{S}_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{S}_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \hat{S}_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

- (a) Calculate the commutator $[\hat{S}_y, \hat{S}_z]$ and express your answer in terms of $\hat{S}_x$.

[5 marks]

- (b) Calculate the uncertainty of S_x in the state $|W\rangle = \begin{pmatrix} (1+i)/2 \\ 1/\sqrt{2} \end{pmatrix}$.

[10 marks]

- (c) A measurement of the y -component of the spin (S_y) yields a negative value. What is the state of the system immediately after the measurement?

[11 marks]

- (d) Consider the time-dependent wavefunction

$$|\phi(t)\rangle = \begin{pmatrix} \alpha e^{-i\beta t} \\ \sqrt{2}\alpha \end{pmatrix}$$

where α and β are positive real constants.

Calculate α so that $|\phi(t)\rangle$ is normalized.

Calculate and plot $\langle \hat{S}_z \rangle$ as a function of time in the state $|\phi(t)\rangle$.

[10 marks]

3. Consider a particle of mass m on an infinite one-dimensional line. It is subject to the potential

$$V(x) = \begin{cases} 0 & x < 0 & \text{(region 1)} \\ 4V_0 & 0 < x < L & \text{(region 2)} \\ V_0 & x > L & \text{(region 3)} \end{cases}$$

where V_0 is a positive constant. For $V_0 < E < 4V_0$, the time-independent Schroedinger equation has a solution with energy E , of the form

$$\begin{aligned} \psi(x) &= \psi_1(x) = e^{ik_1x} + Ae^{-ik_1x} && \text{(region 1)} \\ \psi(x) &= \psi_2(x) = Be^{\alpha x} + Ce^{-\alpha x} && \text{(region 2)} \\ \psi(x) &= \psi_3(x) = De^{ik_3x} && \text{(region 3)} \end{aligned}$$

where k_1 , k_3 and α are real and positive.

- (a) Express the constants k_1 , k_3 and α in terms of E and V_0 .
[8 marks]
- (b) The solution written above can be interpreted as a scattering situation. Identify the terms representing incident, reflected and transmitted waves.
[3 marks]
- (c) Define the reflection probability (R) and transmission probability (T) in terms of the constants appearing in the wavefunction.
[5 marks]
- (d) What are the boundary conditions that the solution must satisfy at $x = 0$ and at $x = L$? Use these boundary conditions to write four equations for the constants A , B , C , D .
[8 marks]
- (e) How does the form of the solution change when $E > 4V_0$?
How does the form of the solution change when $E < V_0$?
[8 marks]

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PARTIAL ANSWERS
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1. Question 1.

Consider a particle of mass m on an infinite one-dimensional line.

(a) Question 1(a)

If the position operator $\hat{x}$ is applied to a wavefunction $\phi(x)$, what is the result?

If the momentum operator $\hat{p}$ is applied to $\phi(x)$, what is the result?

[3 marks]

[Sample Answer:]

$$\hat{x}\phi(x) = x\phi(x) \qquad \hat{p}\phi(x) = -i\hbar \frac{d}{dx}\phi(x) = -i\hbar\phi'(x)$$

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(b) Question 1(b)

Find out if the functions e^{ikx} and $\sin(kx)$ are eigenfunctions of the momentum operator. If so, find the corresponding eigenvalues.

[6 marks]

[Sample Answer:]

e^{ikx} an eigenfunction??

$$\begin{aligned} \hat{p}e^{ikx} &= -i\hbar\partial_x e^{ikx} = -i\hbar(ik)e^{ikx} = \hbar k e^{ikx} \\ &\implies \text{Yes, eigenfunction, with eigenvalue } \hbar k. \end{aligned}$$

$\sin(kx)$ an eigenfunction??

$$\begin{aligned} \hat{p}\sin(kx) &= -i\hbar\partial_x \sin(kx) = -i\hbar k \cos(kx) \\ &\implies \text{Not an eigenfunction.} \end{aligned}$$

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(c) Question 1(c)

Consider a normalized wavefunction $\phi_0(x)$. The expectation value of momentum in this state is p_0 . Find the expectation value of momentum in the state

$$\phi_{\text{new}}(x) = e^{iqx/\hbar} \phi_0(x).$$

Based on your result, explain the physical significance of multiplying a wavefunction by the phase factor $e^{iqx/\hbar}$.

[9 marks]

[Sample Answer:]

We are given

$$\langle \hat{p} \rangle_{\phi_0} = \langle \phi_0 | \hat{p} | \phi_0 \rangle = p_0$$

We want to find the expectation value in the state $\phi_{\text{new}}(x)$:

$$\begin{aligned} \langle \hat{p} \rangle_{\phi_{\text{new}}} &= \langle \phi_{\text{new}} | \hat{p} | \phi_{\text{new}} \rangle = \int_{-\infty}^{\infty} \phi_{\text{new}}(x)^* \hat{p} \phi_{\text{new}}(x) dx \\ &= -i\hbar \int_{-\infty}^{\infty} \phi_{\text{new}}(x)^* \frac{d}{dx} \phi_{\text{new}}(x) dx \end{aligned}$$

Now

$$\frac{d}{dx} \phi_{\text{new}}(x) = \frac{d}{dx} [e^{iqx/\hbar} \phi_0(x)] = \frac{iq}{\hbar} e^{iqx/\hbar} \phi_0(x) + e^{iqx/\hbar} \frac{d}{dx} \phi_0(x)$$

so that

$$\begin{aligned} \phi_{\text{new}}(x)^* \frac{d}{dx} \phi_{\text{new}}(x) &= [e^{-iqx/\hbar} \phi_0(x)^*] \left(\frac{iq}{\hbar} e^{iqx/\hbar} \phi_0(x) + e^{iqx/\hbar} \frac{d}{dx} \phi_0(x) \right) \\ &= \frac{iq}{\hbar} |\phi_0(x)|^2 + \phi_0(x)^* \frac{d}{dx} \phi_0(x) \end{aligned}$$

Thus

$$\begin{aligned} \langle \hat{p} \rangle_{\phi_{\text{new}}} &= -i\hbar \frac{iq}{\hbar} \int_{-\infty}^{\infty} |\phi_0(x)|^2 dx - i\hbar \int_{-\infty}^{\infty} \phi_0(x)^* \frac{d}{dx} \phi_0(x) dx \\ &= q + \int_{-\infty}^{\infty} \phi_0(x)^* \hat{p} \phi_0(x) dx = q + \langle \hat{p} \rangle_{\phi_0} = q + p_0 \end{aligned}$$

Multiplying by this phase factor $e^{iqx/\hbar}$ shifts the momentum by the quantity q , i.e., the wavefunction is ‘boosted’ by momentum q .

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(d) Question 1(d)

Consider the time-independent wavefunction $\psi(x) = Ne^{-|x|/\lambda}$, i.e.,

$$\psi(x) = \begin{cases} Ne^{-x/\lambda} & \text{for } x > 0 \\ Ne^{+x/\lambda} & \text{for } x < 0 \end{cases}$$

where λ and N are real positive constants.

Calculate N in terms of λ so that the wavefunction is normalized.

If the state of the particle is $\psi(x)$, calculate the probability of finding the particle in the region $x \in [-\lambda, \lambda]$.

[14 marks]

[Sample Answer:]

The norm of the wavefunction is

$$\begin{aligned} \int_{-\infty}^{\infty} dx |\psi(x)|^2 &= \int_{-\infty}^0 dx |\psi(x)|^2 + \int_0^{\infty} dx |\psi(x)|^2 \\ &= \int_{-\infty}^0 dx |N|^2 e^{2x/\lambda} + \int_0^{\infty} dx |N|^2 e^{-2x/\lambda} \\ &= |N|^2 \frac{\lambda}{2} + |N|^2 \frac{\lambda}{2} = |N|^2 \lambda \end{aligned}$$

Since N is positive and real, this means

$$N = \frac{1}{\sqrt{\lambda}}$$

The probability of finding the particle in the region $x \in [-\lambda, \lambda]$ is

$$\begin{aligned} \int_{-\lambda}^{\lambda} dx |\psi(x)|^2 &= \int_{-\lambda}^0 dx |\psi(x)|^2 + \int_0^{\lambda} dx |\psi(x)|^2 \\ &= 2 \int_0^{\lambda} dx |\psi(x)|^2 = 2 \int_0^{\lambda} dx |N|^2 e^{-2x/\lambda} \end{aligned}$$

because of the symmetry of the wavefunction under $x \rightarrow -x$. Thus the probability is

$$\begin{aligned} 2 \int_0^{\lambda} dx |N|^2 e^{-2x/\lambda} &= 2|N|^2 \times \frac{\lambda}{2} \int_0^2 du e^{-u} \\ &= 2|N|^2 \frac{\lambda}{2} \left(1 - \frac{1}{e^2}\right) = 1 - \frac{1}{e^2} \end{aligned}$$

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2. Question 2.

Consider a spin-1/2 system. The components of the spin are described by the operators

$$\hat{S}_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \hat{S}_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \hat{S}_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

(a) Question 2(a)

Calculate the commutator $[\hat{S}_y, \hat{S}_z]$ and express your answer in terms of $\hat{S}_x$.

[5 marks]

[Sample Answer:]

$$\begin{aligned} [\hat{S}_y, \hat{S}_z] &= \hat{S}_y \hat{S}_z - \hat{S}_z \hat{S}_y \\ &= \frac{\hbar^2}{4} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \frac{\hbar^2}{4} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \\ &= \frac{\hbar^2}{4} \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} - \frac{\hbar^2}{4} \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} = i\hbar \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = i\hbar \hat{S}_x \end{aligned}$$

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(b) Question 2(b)

Calculate the uncertainty of S_x in the state $|W\rangle = \begin{pmatrix} (1+i)/2 \\ 1/\sqrt{2} \end{pmatrix}$.

[10 marks]

[Sample Answer:]

We need to calculate

$$\Delta S_x = \sqrt{\langle S_x^2 \rangle - \langle S_x \rangle^2}$$

In the state $|W\rangle$. The operators are

$$S_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad S_x^2 = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \frac{\hbar^2}{4} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{\hbar^2}{4} \mathbb{I}$$

Therefore the expectation values are

$$\begin{aligned} \langle S_x \rangle &= \langle W | \hat{S}_x | W \rangle = \frac{\hbar}{2} \begin{pmatrix} \frac{1-i}{2} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{1+i}{2} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \\ &= \frac{\hbar}{2} \begin{pmatrix} \frac{1-i}{2} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1+i}{2} \end{pmatrix} \\ &= \frac{\hbar}{2} \left\{ \frac{1-i}{2\sqrt{2}} + \frac{1+i}{2\sqrt{2}} \right\} = \frac{\hbar}{2\sqrt{2}} \end{aligned}$$

and

$$\langle S_x^2 \rangle = \langle W | \hat{S}_x^2 | W \rangle = \langle W | \frac{\hbar^2}{4} \mathbb{I} | W \rangle = \frac{\hbar^2}{4}$$

since the state is normalized.

The uncertainty is therefore

$$\Delta S_x = \sqrt{\langle S_x^2 \rangle - \langle S_x \rangle^2} = \sqrt{\frac{\hbar^2}{4} - \left(\frac{\hbar}{2\sqrt{2}} \right)^2} = \frac{\hbar}{2\sqrt{2}}$$

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(c) Question 2(c)

A measurement of the y -component of the spin (S_y) yields a negative value. What is the state of the system immediately after the measurement?

[11 marks]

[Sample Answer:]

The possible values that can be obtained in a measurement are the eigenvalues of the operator, and the state after the measurement is

the eigenstate corresponding to the eigenvalue that is found. Thus, we need to first find the eigenvalues of the operator

$$\hat{S}_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

and then find the eigenstate corresponding to the negative eigenvalue. Finding the eigenvalue: this can be done in couple of different ways. If done correctly, the eigenvalues will be $\pm\hbar/2$. These are the possible values that can be obtained in a measurement of S_y .

A possible calculation of the eigenvalues: an eigenvalue λ of $\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ must satisfy

$$\begin{aligned} \left| \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} - \lambda I \right| &= 0 \\ \implies \left| \begin{pmatrix} -\lambda & -i \\ i & -\lambda \end{pmatrix} \right| &= 0 \\ \implies \lambda^2 - 1 &= 0 \implies \lambda = \pm 1 \\ \implies \text{The eigenvalues of } \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} &\text{ are } \pm 1 \\ \implies \text{The eigenvalues of } \hat{S}_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} &\text{ are } \pm\hbar/2. \end{aligned}$$

The result of the measurement is negative, i.e., equal to $-\hbar/2$. After the measurement, the system wavefunction ‘collapses’ to the eigenstate corresponding to the eigenvalue $-\hbar/2$. So, we need to calculate the eigenstate corresponding to the eigenvalue $-\hbar/2$.

Calling this eigenstate $\begin{pmatrix} \beta \\ \gamma \end{pmatrix}$, we obtain

$$\begin{aligned} \hat{S}_y \begin{pmatrix} \beta \\ \gamma \end{pmatrix} &= -\frac{\hbar}{2} \begin{pmatrix} \beta \\ \gamma \end{pmatrix} \\ \implies \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} \beta \\ \gamma \end{pmatrix} &= -\frac{\hbar}{2} \begin{pmatrix} \beta \\ \gamma \end{pmatrix} \\ \implies \begin{pmatrix} -i\gamma \\ i\beta \end{pmatrix} &= -\begin{pmatrix} \beta \\ \gamma \end{pmatrix} \end{aligned}$$

which gives $\gamma = -i\beta$, so that the eigenstate is

$$\begin{pmatrix} \beta \\ -i\beta \end{pmatrix}$$

The value of β can be calculated by normalization:

$$\beta = \frac{1}{\sqrt{2}} e^{i\chi}; \quad \chi \text{ is an arbitrary real number}$$

Thus the state after the measurement is

$$\frac{1}{\sqrt{2}} e^{i\chi} \begin{pmatrix} 1 \\ -i \end{pmatrix} \quad \text{or} \quad \frac{1}{\sqrt{2}} e^{i\chi} \begin{pmatrix} i \\ 1 \end{pmatrix}$$

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(d) Question 2(d)

Consider the time-dependent wavefunction

$$|\phi(t)\rangle = \begin{pmatrix} \alpha e^{-i\beta t} \\ \sqrt{2}\alpha \end{pmatrix}$$

where α and β are positive real constants.

Calculate α so that $|\phi(t)\rangle$ is normalized.

Calculate and plot $\langle \hat{S}_z \rangle$ as a function of time in the state $|\phi(t)\rangle$.

[10 marks]

[Sample Answer:]

The norm of $|\phi(t)\rangle$ is

$$\langle \phi(t) | \phi(t) \rangle = |\alpha e^{-i\beta t}|^2 + |\sqrt{2}\alpha|^2 = |\alpha|^2 + 2|\alpha|^2 = 3|\alpha|^2$$

Since β is real, we have used $|e^{-i\beta t}|^2 = 1$ above. Since α is also real, we have $\langle \phi(t) | \phi(t) \rangle = 3\alpha^2$. Setting $\langle \phi(t) | \phi(t) \rangle = 1$ gives

$$\alpha = \frac{1}{\sqrt{3}}$$

Thus

$$|\phi(t)\rangle = \begin{pmatrix} \frac{1}{\sqrt{3}} e^{-i\beta t} \\ \frac{\sqrt{2}}{\sqrt{3}} \end{pmatrix} \quad \text{and} \quad \langle \phi(t) | = \left(\frac{1}{\sqrt{3}} e^{+i\beta t} \quad \frac{\sqrt{2}}{\sqrt{3}} \right)$$

Therefore

$$\begin{aligned}
 \langle \hat{S}_z \rangle &= \langle \phi(t) | \hat{S}_z | \phi(t) \rangle \\
 &= \begin{pmatrix} \frac{1}{\sqrt{3}} e^{+i\beta t} & \frac{\sqrt{2}}{\sqrt{3}} \end{pmatrix} \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{3}} e^{-i\beta t} \\ \frac{\sqrt{2}}{\sqrt{3}} \end{pmatrix} \\
 &= \begin{pmatrix} \frac{1}{\sqrt{3}} e^{+i\beta t} & \frac{\sqrt{2}}{\sqrt{3}} \end{pmatrix} \begin{pmatrix} \frac{\hbar}{2\sqrt{3}} e^{-i\beta t} \\ -\frac{\sqrt{2}\hbar}{2\sqrt{3}} \end{pmatrix} = \frac{\hbar}{6} - \frac{1\hbar}{3} = -\frac{\hbar}{6}
 \end{aligned}$$

The plot should be a horizontal straight line at a negative value.

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3. Question 3.

Consider a particle of mass m on an infinite one-dimensional line. It is subject to the potential

$$V(x) = \begin{cases} 0 & x < 0 & (\text{region 1}) \\ 4V_0 & 0 < x < L & (\text{region 2}) \\ V_0 & x > L & (\text{region 3}) \end{cases}$$

where V_0 is a positive constant. For $V_0 < E < 4V_0$, the time-independent Schroedinger equation has a solution with energy E , of the form

$$\begin{aligned}
 \psi(x) &= \psi_1(x) = e^{ik_1x} + Ae^{-ik_1x} & (\text{region 1}) \\
 \psi(x) &= \psi_2(x) = Be^{\alpha x} + Ce^{-\alpha x} & (\text{region 2}) \\
 \psi(x) &= \psi_3(x) = De^{ik_3x} & (\text{region 3})
 \end{aligned}$$

where k_1 , k_3 and α are real and positive.

(a) Question 3(a)

Express the constants k_1 , k_3 and α in terms of E and V_0 .

[8 marks]

[Sample Answer:]

$$k_1 = \sqrt{\frac{2mE}{\hbar^2}}; \quad k_3 = \sqrt{\frac{2m(E - V_0)}{\hbar^2}}; \quad \alpha = \sqrt{\frac{2m(4V_0 - E)}{\hbar^2}}$$

$$- - - - - = * - - - - -$$

(b) Question 3(b)

The solution written above can be interpreted as a scattering situation. Identify the terms representing incident, reflected and transmitted waves.

[3 marks]

[Sample Answer:]

e^{ik_1x} $\longrightarrow$ incident wave

Ae^{-ik_1x} $\longrightarrow$ reflected wave

De^{ik_3x} $\longrightarrow$ transmitted wave

$$- - - - - = * - - - - -$$

(c) Question 3(c)

Define the reflection probability (R) and transmission probability (T) in terms of the constants appearing in the wavefunction.

[5 marks]

[Sample Answer:]

The reflection probability is the reflected current divided by the incident current.

The transmission probability is the transmitted current divided by the incident current.

$$\begin{aligned} \text{incident current} &= 1 \times \frac{\hbar k_1}{m} \\ \text{reflected current} &= |A|^2 \times \frac{\hbar k_1}{m} \\ \text{transmitted current} &= |D|^2 \times \frac{\hbar k_3}{m} \end{aligned}$$

Thus

$$R = |A|^2 \quad T = |D|^2 \frac{k_3}{k_1}$$

$$- - - - - = * - - - - -$$

(d) Question 3(d)

What are the boundary conditions that the solution must satisfy at $x = 0$ and at $x = L$? Use these boundary conditions to write four equations for the constants A, B, C, D .

[8 marks]**[Sample Answer:]**

Boundary conditions: the wavefunction and its derivative must be continuous at $x = 0$ and at $x = L$. Thus four boundary conditions:

$$\psi_1(0) = \psi_2(0); \quad \psi'_1(0) = \psi'_2(0); \quad \psi_2(L) = \psi_3(L); \quad \psi'_2(L) = \psi'_3(L)$$

Each gives a relation between A, B, C, D . The four equations are

$$\begin{aligned} 1 + A &= B + C \\ ik_1 - ik_1A &= \alpha B - \alpha C \\ Be^{\alpha L} + Ce^{-\alpha L} &= De^{ik_3L} \\ \alpha Be^{\alpha L} - \alpha Ce^{\alpha L} &= ik_3De^{ik_3L} \end{aligned}$$

This is a system of linear equations for the four constants. In matrix form

$$\begin{pmatrix} 1 & -1 & -1 & 0 \\ -ik_1 & -1 & 1 & 0 \\ 0 & e^{\alpha L} & e^{-\alpha L} & -e^{ik_3L} \\ 0 & \alpha e^{\alpha L} & -\alpha e^{-\alpha L} & -ik_3e^{ik_3L} \end{pmatrix} \begin{pmatrix} A \\ B \\ C \\ D \end{pmatrix} = \begin{pmatrix} -1 \\ -ik_1 \\ 0 \\ 0 \end{pmatrix}$$

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(e) Question 3(e)

How does the form of the solution change when $E > 4V_0$?

How does the form of the solution change when $E < V_0$?

[8 marks]**[Sample Answer:]**

For $E > 4V_0$, the solutions in regions 1 and 3 have the same form, but in region 2 we get plane wave solutions instead of exponential solutions:

$$\psi_2(x) = Ee^{ik_2x} + Fe^{-ik_2x}$$

with $k_2 = \sqrt{2m(E - 4V_0)/\hbar^2}$.

For $E < V_0$, there is no transmission into region 3. The wavefunctions in regions 1 and 2 can have the same form but in region 3 we would have an exponentially decaying function:

$$\psi_3(x) = Ge^{-\gamma x}$$

with $\gamma = \sqrt{2m(V_0 - E)/\hbar^2}$.

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